

EE610: Analog IC Design

Design Project - 2 Report

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Roll no.	R_L	C_L	Input Stage
251040092	10 k Ω	2 pF	nMOS

Complete Opamp Schematic and Architecture

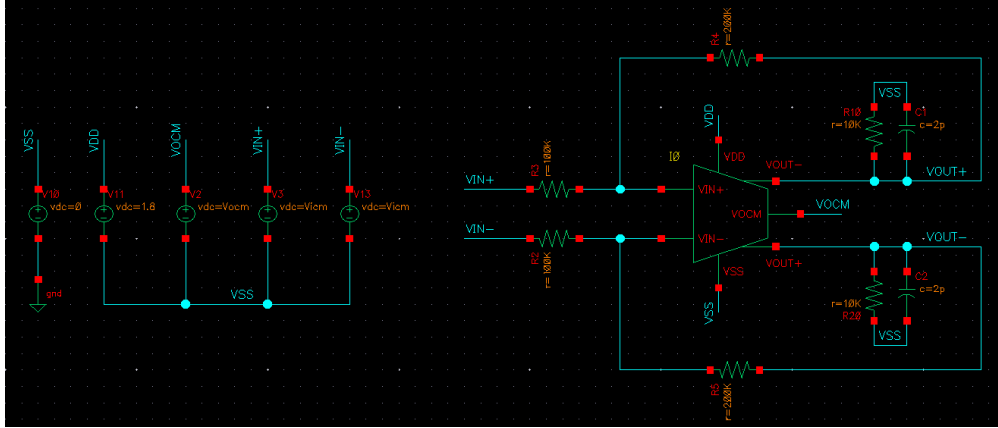


Figure 1: Fully-differential two-stage Miller opamp in -2 inverting configuration

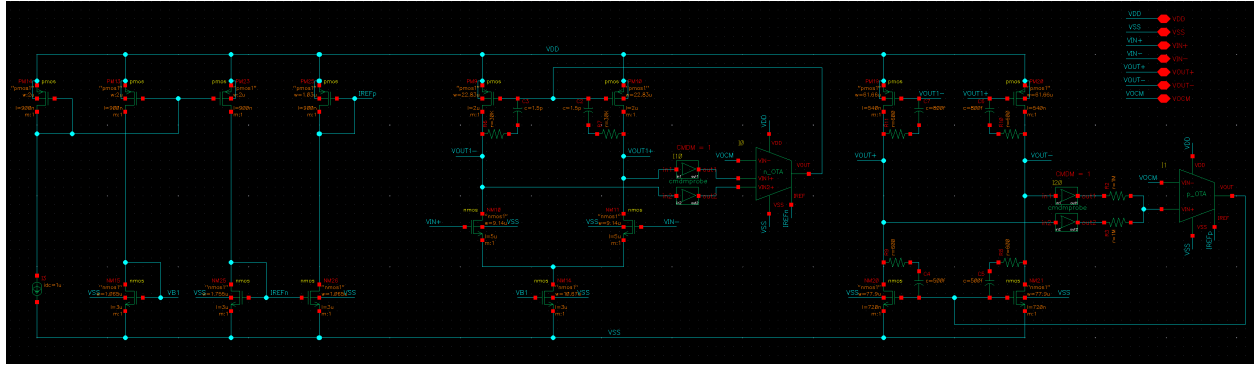
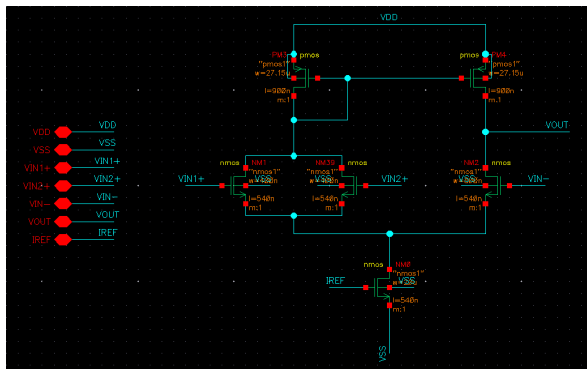
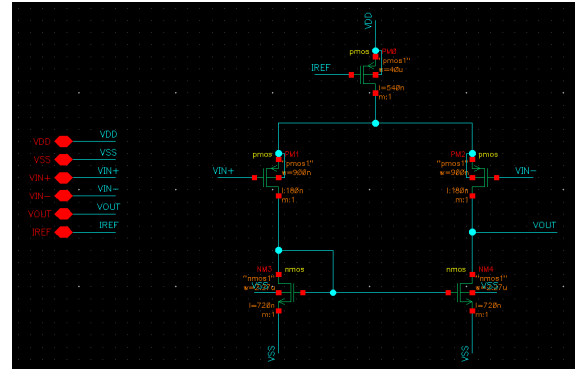


Figure 2: Fully-differential Miller compensated two-stage opamp with CMFB circuits



(a) CMFB1: nMOS input 5-transistor OTA for first stage common-mode control



(b) CMFB2: pMOS input 5-transistor OTA for second stage common-mode control

Architecture Description: The designed operational amplifier employs a fully-differential two-stage topology with Miller compensation and Common-Mode Feedback (CMFB) circuits.

- **First Stage:** nMOS differential pair (NM10, NM11) with pMOS active loads (PM9, PM10).

- **Second Stage:** pMOS differential pair (PM19, PM20) with nMOS active loads (NM20, NM21).
- **Biasing Circuit:** Current mirrors derived from $1\mu A$ reference current source.
- **CMFB Circuits:** Two CMFB loops for each stage using 5-transistor OTA topology.
- **Compensation:** Miller compensation with resistors for each CMFB stage and differential stage.

Design Methodology

Specifications Analysis

The opamp targets a closed-loop gain of -2 ($R_1 = 100k\Omega$, $R_2 = 200k\Omega$, $\beta = 1/3$) with DC loop gain ≥ 60 dB, driving $R_L = 10k\Omega \parallel C_L = 2pF$. Key requirements: closed-loop 3-dB bandwidth ≥ 1 MHz (no peaking), CMFB loops with $\geq 60^\circ$ phase margin and UGB $\geq 1/4 \times$ differential UGB, second-stage $V_{cm} = V_{dd}/2$, and all CMFB resistors $< 1M\Omega$.

Gain Distribution

Target loop gain: $20 \log(A_1 \times A_2 \times \beta) \geq 60$ dB

$A_1 + A_2 \geq 60 - 20 \log(1/3) \approx 69.54$ dB

Chosen distribution: $A_1 \approx 50$ dB, $A_2 \geq 20$ dB

First Stage Design

$$A_1 = g_m(r_{on} \parallel r_{op}) \approx 350$$

$$r_{on} = r_{op} = r_o \Rightarrow g_m r_o \approx 700$$

NM10, NM11 (Input pair):

From g_m/g_{ds} curves: $L = 5\mu m$, $V_{ov} = 100.129mV$

$$g_m/g_{ds} = 715.41, \quad I_D/W = 0.547$$

$$I_D = 5\mu A \Rightarrow W = 9.14\mu m$$

$$g_m/I_D = 13.028652 \Rightarrow g_m = 65.14\mu S$$

PM9, PM10 (Active load):

$$g_{ds} = 0.1665\mu S, \quad I_d = 5\mu A, \quad g_{ds}/I_D = 0.03329$$

From curves: $L = 2\mu m$, $V_{ov} = 96.125mV$

$$I_D/W = 0.219 \Rightarrow W = \frac{5\mu A}{0.219} = 22.83\mu m$$

$$g_m/I_D = 13.22377 \Rightarrow g_m = 66.12\mu S$$

NM14 (Tail current):

$$I_D = 2 \times 5\mu A = 10\mu A$$

From curves: $L = 3\mu m$, $V_{ov} = 100.092mV$

$$I_D/W = 0.937 \Rightarrow W = 10.67\mu m$$

- **NM10, NM11 (Input pair):** $W = 9.14\mu m$, $L = 5\mu m$, $I_D = 5\mu A$, $g_m = 65.14\mu S$
- **PM9, PM10 (Active load):** $W = 22.83\mu m$, $L = 2\mu m$, $I_D = 5\mu A$, $g_m = 66.12\mu S$
- **NM14 (Tail current):** $W = 10.67\mu m$, $L = 3\mu m$, $I_D = 10\mu A$

Second Stage Design

$$A_2 = g_m(r_{on} \parallel r_{op} \parallel R_L) \geq 10$$

$$\text{With } R_L = 10k\Omega, \quad r_{on} = r_{op} = r_o \approx 30k\Omega \Rightarrow g_m \geq 1.17mS, \quad g_{ds} \approx 50mS$$

PM19, PM20 (Input pair):

$$\text{From curves: } L = 540nm, \quad V_{ov} = 399.906mV$$

$$g_m/g_{ds} = 68.21, \quad g_{ds}/I_D = 0.064404 \Rightarrow I_D = 776.35\mu A$$

$$I_D/W = 8.395 \Rightarrow W = 61.66\mu m$$

NM20, NM21 (Active load):

$$g_{ds}/I_D = 0.064405 \text{ (matched to input pair)}$$

$$\text{From curves: } L = 720nm, \quad V_{ov} = 150.612mV$$

$$I_D/W = 6.645 \Rightarrow W = 77.90\mu m$$

- **PM19, PM20 (Input pair):** $W = 61.66\mu m$, $L = 540nm$, $g_m = 2.8mS$
- **NM20, NM21 (Active load):** $W = 77.90\mu m$, $L = 720nm$

Biasing and CMFB Circuits Design

All bias currents are derived from the $1\mu A$ reference via width scaling in current mirrors, with CMFB bias voltages similarly generated to maintain saturation. CMFB1 uses a modified nMOS-input 5T OTA (split input pair from Project 1), while CMFB2 employs a pMOS-input 5T OTA.

Compensation Network Design

CMFB1 Stability:

$$R_{6,7} \geq \frac{1}{2g_{m(PM10,PM11)}} = \frac{1}{2 \times 65.95\mu S} = 7.58k\Omega$$

$$\text{Chose } R_{6,7} = 30k\Omega, \quad \text{Through simulation sweep: } C_{2,3} = 1.5pF \Rightarrow PM = 68^\circ$$

CMFB2 Stability:

$$R_{8,9} \geq \frac{1}{2g_{m(NM20,NM21)}} = \frac{1}{2 \times 6.5mS} = 76.9\Omega$$

$$\text{Chose } R_{8,9} = 600\Omega, \quad \text{Through simulation sweep: } C_{4,5} = 500fF \Rightarrow PM = 64^\circ$$

Differential Loop Stability:

$$R_{10,11} \geq \frac{1}{2g_{m(PM19,PM20)}} = \frac{1}{2 \times 2.8mS} = 179.9\Omega$$

Chose $R_{10,11} = 600\Omega$, Through simulation sweep: $C_{6,7} = 800fF \Rightarrow PM = 68^\circ$

Final Compensation Components:

- **CMFB1:** $R = 30k\Omega$, $C = 1.5pF$, $PM = 68^\circ$
- **CMFB2:** $R = 600\Omega$, $C = 500fF$, $PM = 64^\circ$
- **Differential:** $R = 600\Omega$, $C = 800fF$, $PM = 68^\circ$

Block-level Design Summary

Table 1: Block-level Design Summary

Parameter	First Stage	Second Stage	Main Diff. Loop	CMFB1	CMFB2
G_m	65.14 μS	2.8 mS	-	-	-
I_d	5 μA	776.35 μA	-	-	-
C_c	-	-	800 fF	1.5 pF	500 fF
R_c	-	-	600 Ω	30 k Ω	600 Ω
PM	-	-	68 $^\circ$	68 $^\circ$	64 $^\circ$

Result #1: Simulated DC Operating Point

Table 2: DC Operating Point Results

Parameter	Designed	Simulated	Units
First Stage g_m	65.14	65.9537u	μS
First Stage I_D	5	5.07119	μA
Second Stage g_m	2800	2778.52	μS
Second Stage I_D	776.35	727.061	μA

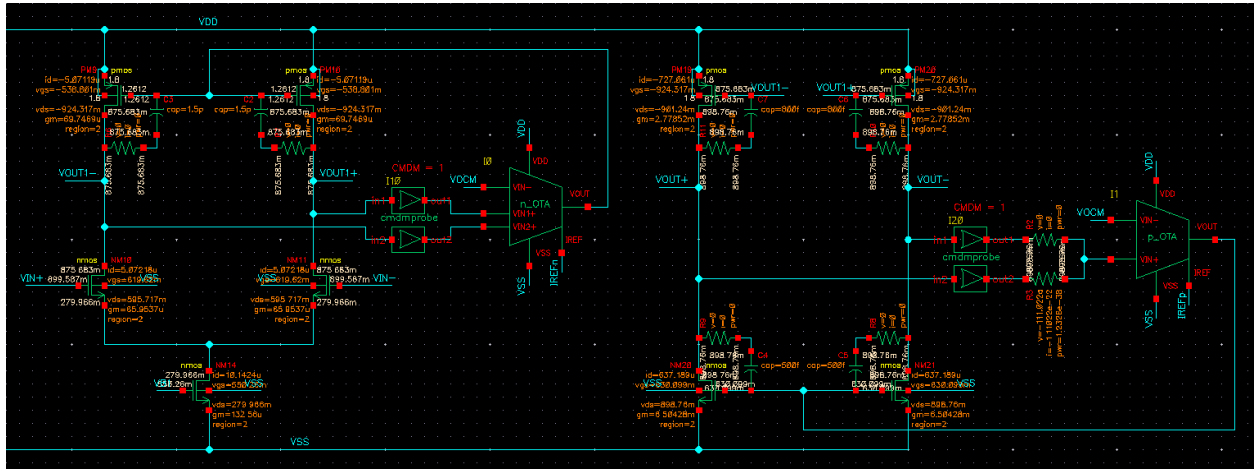


Figure 4: DC operating point simulation

Result #2: Differential-Mode Loop Gain

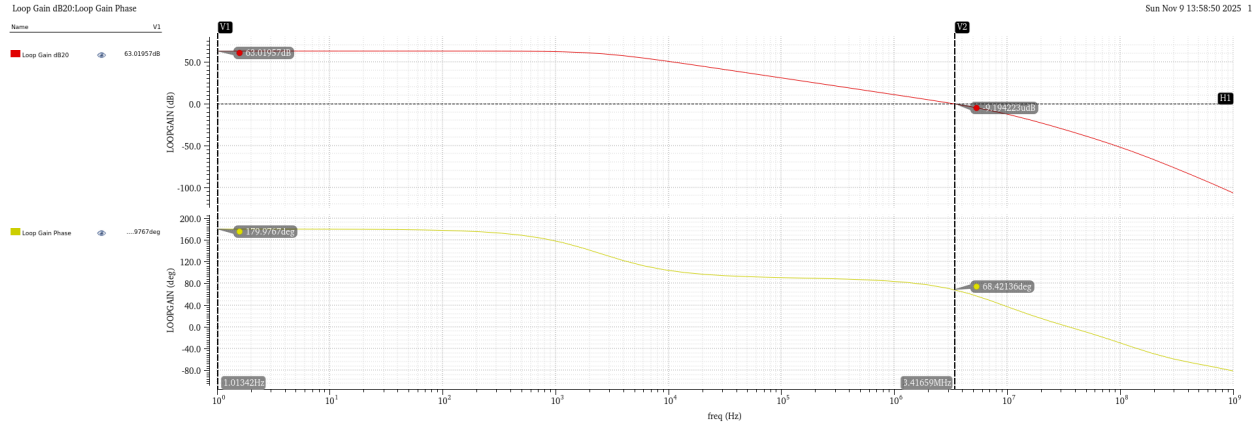


Figure 5: Differential-mode loop gain mag and phase response UGF: 3.42MHz, PM: 68.42deg

Result #3: Closed-loop Frequency Response

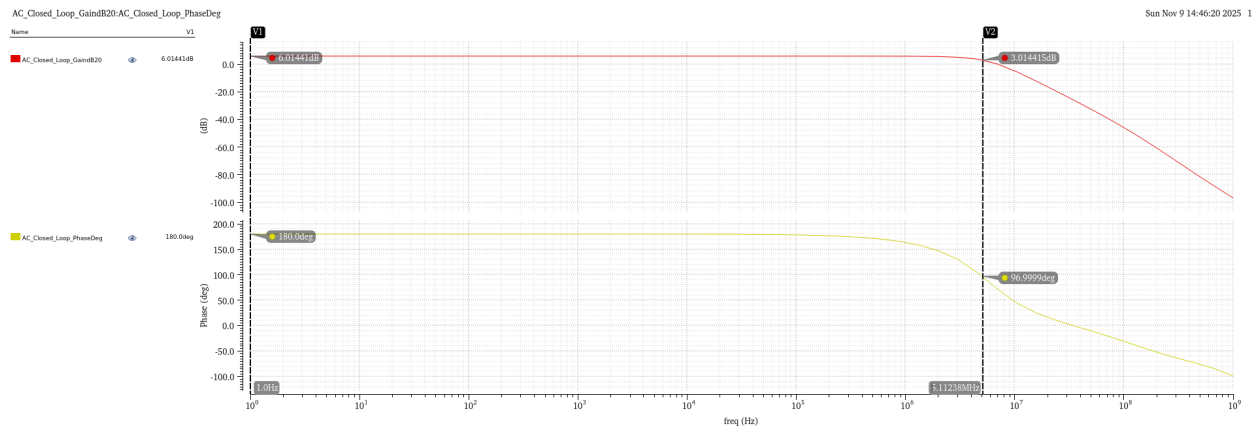


Figure 6: Closed-loop frequency response -3dB Bandwidth: 5.11MHz, No peaking observed

Result #4: CMFB Loop Gains

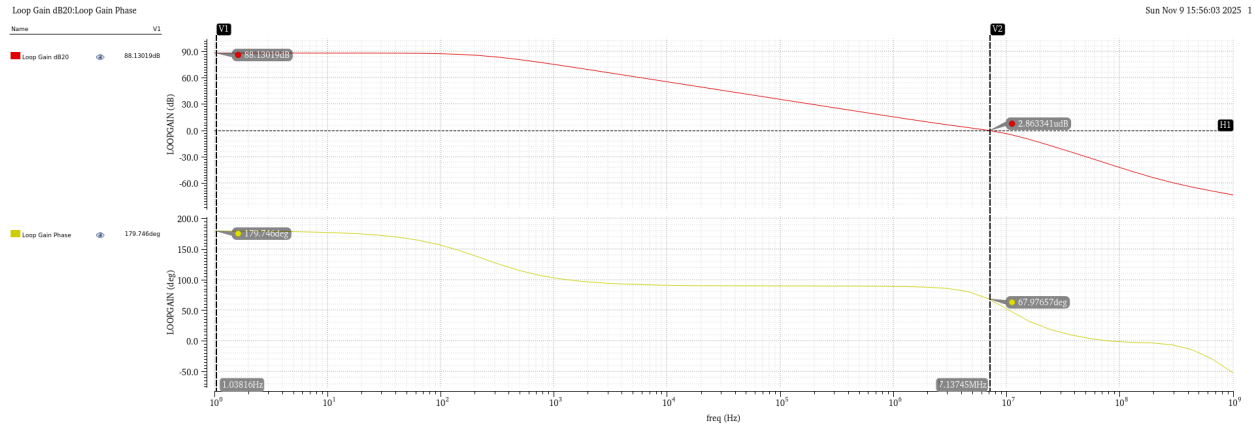


Figure 7: CMFB1 Loop Gain UGF: 7.13745MHz, PM: 67.97657deg

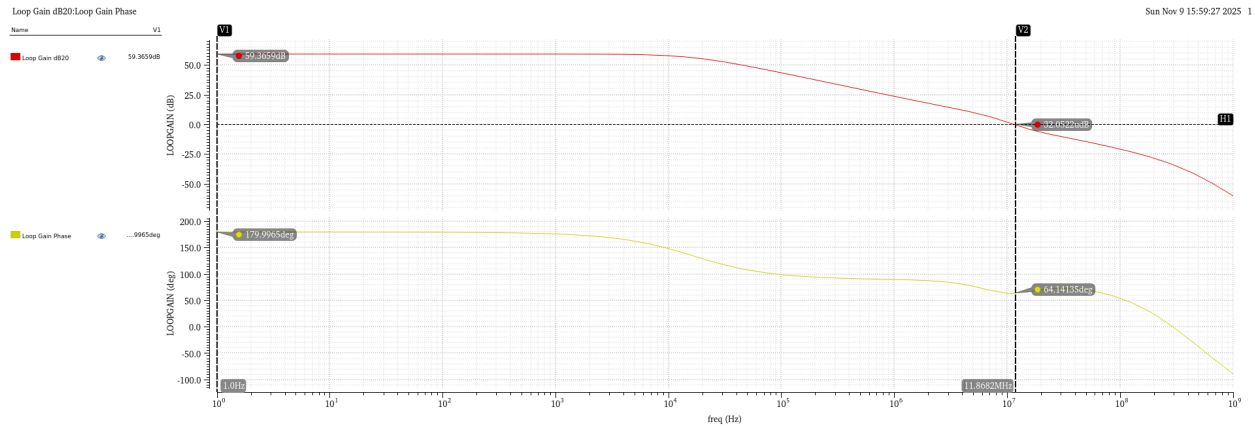


Figure 8: CMFB2 Loop Gain UGF: 11.8682MHz, PM: 64.14135deg

Result #5: Differential Step Response

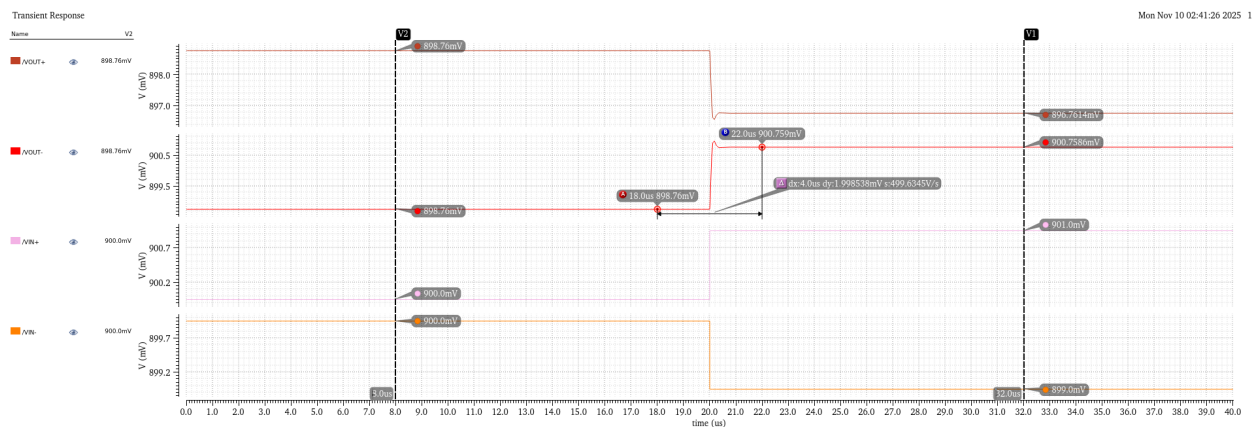


Figure 9: Differential output response for 1mV differential step input

Result #6: Common-mode Step Response

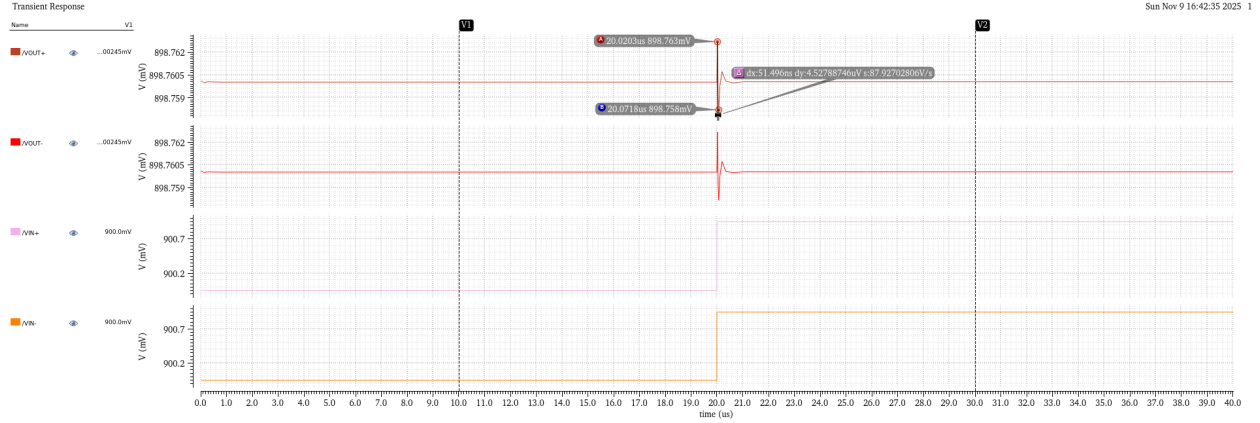


Figure 10: Output response for 1mV common-mode step input

Explanation: The negligible output change confirms excellent common-mode rejection ($A_{CM} \approx 0$), matching theoretical expectations. This results from the tail current source (NM14) maintaining constant current, forcing the differential pair source to follow common-mode inputs and creating balanced current changes.